

SHEN YINGJUN (SIMON)

Email: jiucaai@yonsei.ac.kr | Phone: +82 10-2713-1677 | GitHub: github.com/jiucaai233 | Website: jiucaai233.github.io/Yingjun-s-website/

RESEARCH INTERESTS

Skill-based and hierarchical reinforcement learning, model-based RL for long-horizon manipulation, and compositional generalization of learned skills. Currently investigating why pretrained skill priors fail to recombine zero-shot, through a first-author diagnostic working paper on SkiMo (Franka Kitchen).

EDUCATION

Yonsei University

Seoul, South Korea

B.E. in Applied Information Engineering GPA: 3.83/4.3 (96.0/100)

Sep. 2022 – Aug. 2027

Selected Coursework: Introduction to Artificial Intelligence (A0), Network Data Analysis (A0), Advanced Data Analytics (A0), Natural Language Information Analysis (A+), Time Series Analysis (B+)

Independent Study: UC Berkeley CS285 (Deep Reinforcement Learning) — completed the full curriculum with from-scratch implementations of core algorithms (policy gradients, actor-critic, SAC, model-based RL, offline RL). [GitHub] **Beijing Foreign Studies University (BFSU) International School** Beijing,

China

High School Diploma

Sep. 2019 – Jun. 2022

RESEARCH EXPERIENCE

Agent-OS Lab, Yonsei University

Apr. 2026 – Present

Research Intern

Seoul, South Korea

- **Independent research (first-author working paper):** “*Order Fixation in Skill-Based RL: Why Learned Skills Refuse to Recombine*” — designed a four-probe diagnostic protocol on SkiMo (Franka Kitchen) to localize the failure of zero-shot skill recombination; buffer-transplant and relabeling ablations indicate the critic encodes trained transition order rather than outcome-sensitive value, while the world model and skill execution remain intact. Preparing for conference submission.
- **Vision-Grounded Speech Repair (VGSR, group project):** Contributing to a framework that mitigates STT errors in robotic task execution via object-graph and spatial relation inference; co-drafting a research paper.

DSL (Yonsei Data Science Lab)

Mar. 2024 – Mar. 2025

Research Member

Seoul, South Korea

- Developed conditional generative models for image synthesis, fine-tuned YOLO detectors, and applied deep learning to medical image segmentation.

ENGINEERING EXPERIENCE

YBIGTA (Yonsei Big Data Society)

Dec. 2025 – Present

Core Member, Data Engineering Team

Seoul, South Korea

- **ROBO404++ (Robotic Control Pipeline):** Built an autonomous navigation simulation trained with Soft Actor-Critic (SAC); integrated YOLO (ONNX) perception into a real-time perception-control loop in ROS.
- **Transparent-Audit System:** Designed a FastAPI backend with LLM integration for receipt parsing and anomaly detection; deployed containerized services with Docker/Kubernetes.

SKILLS

Research: Python, PyTorch (primary), NumPy; RL codebases (SPiRL, SkiMo, SAC); MuJoCo

Engineering: Docker, Kubernetes, ROS, Gazebo, Git, AWS, FastAPI, Next.js/React

Languages: Chinese (native), English (full professional proficiency; B.E. coursework conducted in English), Korean (conversational; completed Yonsei Korean language coursework)

HONORS & AWARDS

1st Place, YBIGTA 26-1 Conference Project (for ROBO404++)

Jun. 2026

Special Award, 2026 SKYSH Hackathon (SNU–Korea–Yonsei–SKKU–Hanyang coalition)

2026

Yonsei University Merit-Based Scholarship

2023 – 2025

GLC–SeAh Holdings Scholarship (leadership and engineering excellence)

2024

GLC Distinction Award (academic excellence, Global Leaders College)

2024